

Xiwen (Francis) Chen

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EDUCATION

University of Southern California

M.S. in Computer Science, Artificial Intelligence Track; GPA: 3.74 / 4.00

Jan. 2024 – May 2026

Los Angeles, CA

South China Normal University

B.E. in Software Engineering; GPA: 88.7 / 100

Sep. 2019 – Jul. 2023

Guangzhou, China

TECHNICAL SKILLS

Languages: Python, C++, SQL, Bash, JavaScript

ML Frameworks & Tools: PyTorch, scikit-learn, MONAI, NumPy, pandas, SimpleITK, CUDA, Docker, Git, Linux, Slurm

Research Areas: Data-Centric AI, Agentic AI, Domain Adaptation/Generalization, Active Learning, RL, Optimization

EXPERIENCE

University of Southern California

Apr. 2024 – May 2026

Research Assistant — Data-Centric AI & Training Data Curation — Advisor: Prof. Ruishan Liu

Los Angeles, CA

- Led two first-author projects on data-efficient cross-domain learning, curating compact target-compatible training subsets to reduce source-data usage while improving adaptation under label scarcity and distribution shift.
- Designed TAVO, a target-aware data valuation framework that addresses unstable single-criterion source selection by fusing 8 feature/gradient-based valuation signals through surrogate-guided optimization under fixed training budgets.
- Developed TAC, a source–target curation method that couples active target acquisition with source selection, using queried target labels as reliability anchors for compact, target-aligned source subset construction.
- Built reproducible PyTorch pipelines for data-curation experiments across segmentation and classification tasks, covering source–target split construction, subset-selection baselines, multi-budget training, metric aggregation, ablation studies, and EfficientViT/nnU-Net backbone validation.
- Authored first-author manuscripts under NeurIPS/AAAI review; TAVO and TAC improved average target-domain Dice by +0.02 over strong domain-adaptation and data-curation baselines, and by +0.04 over naive full-source training.

Lab for ML, Health and Biomedicine, USC

May 2026 – Present

Research Assistant — Cost-Aware Agentic Planning for Radiotherapy

Los Angeles, CA

- Leading RadOnc-Gym, a cost-aware LLM/RL planning environment for radiotherapy, where agents iteratively adjust clinical objectives while reducing expensive Gurobi re-optimization and DVH/clinical-score verification calls.
- Extended the OpenKBP-Opt/Gurobi pipeline into a configurable optimization simulator, exposing 12 tunable objective weights and optimization-mode settings for Gurobi-LP fluence optimization, plan-dose generation, and DVH/clinical scoring.
- Built a Gym-style LLM/RL environment that wraps the Gurobi simulator into a state–action–reward loop, where agents observe DVH/clinical feedback, adjust objective weights or optimization modes, and trade off fast surrogate scoring against full-resolution verification costs.
- Improved plan quality with fewer full-resolution Gurobi calls than fixed-verification and always-full-solve baselines by combining surrogate-guided search with selective verifier use.

Sitech Information Technology Co., Ltd.

Mar. 2023 – Sep. 2023

Research Engineering Intern — Distributed Storage Systems, Erasure Coding

Beijing & Shenzhen, China

- Operationalized CUHK-led network/erasure-coding research for enterprise distributed storage, translating RDP, EVENODD, and circular-shift MDS designs from standalone research prototypes into NCDSS engineering validation workflows.
- Deployed Linux-based NCDSS test clusters and implemented end-to-end validation across coding libraries, demo modules, and client I/O tools, covering encode/decode/repair/update paths under simulated storage failures.
- Produced correctness and performance reports across K=4, M=2 redundancy layouts, 2KB–1GB workloads, and RAID1/RS/CRS/ISA-L/circular-shift baselines to assess performance and deployment trade-offs for enterprise storage systems.

Kingsun Technologies Co., Ltd.

Jun. 2022 – Sep. 2022

AI Engineering Intern — Digital Human Teaching & Speech Synthesis

Shenzhen, China

- Contributed to an educational speech-synthesis system for scalable course-audio generation, developing Tacotron2-based neural TTS workflows with attention/alignment checks to improve text-audio consistency.
- Processed 80+ hours of MOOC speech data using Python/Librosa pipelines for audio cleaning, segmentation, matching, validation, and speaker-adaptation experiments supporting personalized voice generation for digital learning content.

RESEARCH EXPERIENCE

South China Normal University

Sep. 2020 – Apr. 2022

Research Contributor — 4th Author, DOI: 10.3233/JIFS-212511

Guangzhou, China

- Executed CT-sequence preprocessing and PyTorch training pipelines for an 831-case stroke recognition study, including slice selection, resizing to 10×224×224 inputs, median-filter denoising, histogram equalization, and train/validation evaluation.
- Ran ablation experiments across 2D/3D CNN, ResNet, C3D, and SE-attention variants, showing that preprocessing, 3D convolutions, and channel attention improved stroke CT-sequence classification over non-attention 3D baselines.